

OPEN Compute Project

HPM and Peripheral Programmable Device (PD) Specification: Schema Documentation

Part of the Modular Hardware Systems (MHS) Rev 1.0 Family

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L = Local, **R** = Required

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Top-level Properties

Specification: This resource shall represent the hardware and firmware properties of a field programmable gate array or equivalent programmable device used to enable a management interface for an HPM.

Property	Type	L	R	Specification
_comment	string			This property shall contain a general comment about a data file
Copyright	string			This property shall contain a copyright statement for the FRU data. The omission of this property shall not imply that this FRU data is not copyrighted.
Version	string	Y	Y	This property shall contain the current file version for this FRU data.
SchemaVersion	string	Y	Y	This property shall contain the revision of the Programmable Device schema that this FRU data was built against.
GitHubChecksum	string			This property shall contain the GitHub checksum for this FRU data.
Author	string			This property shall contain the author of this FRU data.
General	General	Y	Y	This property shall contain general information about the Programmable Device plus overall build information.
IPBlocks	IPBlocks	Y	Y	This property shall contain information about the IP blocks compiled into this Programmable Device.
MemoryMap	MemoryMap	Y	Y	This property shall contain information about the memory map for this Programmable Device.

Property	Type	L	R	Specification
Registers	Registers			This property shall contain information defining the registers exported by this Programmable Device.
PhysicalSignals	PhysicalSignals	Y	Y	This property shall contain information about the physical signals supported by this Programmable Device.
VirtualWires	VirtualWires	Y	Y	This property shall contain information about the virtual wires supported by this Programmable Device.
Buses	Buses	Y	Y	This property shall contain information about the logical and physical buses connected to this Programmable Device.

General

Specification: This property shall contain general information about the Programmable Device plus overall build information.

Property	Type	L	R	Specification
VendorID	string	Y	Y	This property shall contain the PCIe vendor ID for this Programmable Device.
UniqueID	string	Y	Y	This property shall contain the unique ID specific to this Programmable Device.
Model	string			This property shall contain the model for this Programmable Device, as reported by the manufacturer.
Manufacturer	string			This property shall contain the manufacturer for this Programmable Device.

Property	Type	L	R	Specification
Build	General.Build			This property shall contain information about the firmware build for this Programmable Device.

General.Build

Specification: This property shall contain information about the firmware build for this Programmable Device.

Property	Type	L	R	Specification
Version	string			This property shall contain the version number for the Programmable Device build.
BuildTimestamp	string			This property shall contain the timestamp when the Programmable Device build completed.
Builder	string			This property shall contain the name of individual who performed the build.

IPBlocks

Specification: This property shall contain information about the IP blocks compiled into this Programmable Device.

Array Item Properties:

Property	Type	L	R	Specification
Identifier	NamedComponent	Y	Y	This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components).

Property	Type	L	R	Specification
Alias	ComponentAlias			This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components).
Type	string	Y	Y	This property shall contain the type of this IP block.
Spec	SpecReference			This property shall contain a reference to an OCP specification.
GUID	string	Y	Y	This property shall contain a globally unique identifier for this IP block.
HWID	string	Y	Y	This property shall contain a hardware identifier for this IP block.
SupportedSOCSockets	integer			This property shall contain the maximum number of CPU SOC sockets supported by this IP block.
ExtendedVersion	IPBlocks.ExtendedVersion			This property shall contain extended version information for this IP block.
ConnectedBuses	ReferencedBusList			This object shall contain a collection of references to hardware or logical buses.
OEM	object			This property shall contain OEM-specific information for this IP Block.

IPBlocks.ExtendedVersion

Specification: This property shall contain extended version information for this IP block.

Property	Type	L	R	Specification
Originator	string			This property shall contain the originating entity for this IP block.
OriginalIPVersion	string			This property shall contain the source version for this IP block, prior to modifications.
ModifiedBy	string			This property shall contain the name of the entity that has modified this IP block.
ModificationVersion	string			This property shall contain the modification version for this IP block.
ModificationTimestamp	string			This property shall contain the timestamp for when modifications to this IP block were completed.
ModificationContact	string			This property shall contain contact information for the author of modifications to this IP block, such as an email address.

MemoryMap

Specification: This property shall contain information about the memory map for this Programmable Device.

Property	Type	L	R	Specification
Version	string			This property shall contain the current version for this MemoryMap data.
Regions	array of MemoryMap.Regions	Y	Y	This property shall contain information about about different regions of memory in the map.

MemoryMap.Regions

Specification: This property shall contain information about about different regions of memory in the map.

Property	Type	L	R	Specification
Identifier	NamedComponent	Y	Y	This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components).
Alias	ComponentAlias			This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components).
MappedIPBlock	string	Y	Y	This property shall contain the named identifier for the IP Block in the `IPBlocks` section of this document to which this memory region maps.
Version	string			This property shall contain the version of this IP block for CSRs.
RegionOffsetBytesHex	string	Y	Y	This property shall contain the offset in memory for the beginning of this region, in hexadecimal format.
RegionSizeBytes	integer	Y	Y	This property shall contain the size in bytes for this memory region.
RegisterMap	array of MemoryMap.Regions.RegisterMap	Y	Y	This property shall contain information about about registers in the map.
BAR	array of MemoryMap.Regions.BAR			This property shall contain information about the Base Address Registers (BARs) of this device.

MemoryMap.Regions.RegisterMap

Specification: This property shall contain information about about registers in the map.

Property	Type	L	R	Specification
Identifier	NamedComponent	Y	Y	This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components).
Alias	ComponentAlias			This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components).
Spec	SpecReference	Y	Y	This property shall contain a reference to an OCP specification.
XIOs	integer			This property shall contain the number of XIOs that the register set represents.
PICPWRs	integer			This property shall contain the number of PICPWRs that the register set represents.
OffsetBytes	integer	Y	Y	This property shall contain the offset from the beginning of this region for this set of registers. The initial position for this set of registers may thus be calculated as `RegionOffsetBytes` + `OffsetBytes`.

Property	Type	L	R	Specification
SizeBytes	integer	Y	Y	This property shall contain the size in bytes for this set of registers. If this property is omitted, the set of registers shall be presumed to be one byte in size.
MappedRegister	string			This property shall contain the named identifier for the register in the `Registers` section of this document to which this register maps. This property shall be omitted if the registers map in standard fashion per a specification as indicated by the presence of the `Spec` property.
InactiveRanges	array of MemoryMap.Regions.RegisterMap.InactiveRanges			This property shall detail ranges of registers from the spec not active. This property shall be omitted if `Spec` is not present.
AccessDomain	string			This property shall contain the access domain for this register/register-set. Possible values: `BMCOnly`, `HostOnly`, `Shared`
HostAccessDescription	string			This property shall contain a description of host-visible behavior, required when AccessDomain is HostOnly or Shared.

MemoryMap.Regions.RegisterMap.InactiveRanges

Specification: This property shall detail ranges of registers from the spec not active. This property shall be omitted if `Spec` is not present.

Property	Type	L	R	Specification
Start	integer			This property shall contain the first in a range of registers that was not

Property	Type	L	R	Specification
				implemented.
Width	integer			This property shall contain the number of registers in a range that was not implemented.

MemoryMap.Regions.BAR

Specification: This property shall contain information about the Base Address Registers (BARs) of this device.

Property	Type	L	R	Specification
Index	integer			This property shall contain the Base Address Register index.
BaseAddressHex	string			This property shall contain the base address value advertised to the host, in hexadecimal.
SizeBytes	integer			This property shall contain the size of the BAR window.
Prefetchable	boolean			This property shall indicate whether the BAR is marked prefetchable.
AddressWidth	string			This property shall indicate whether the BAR is 32-bit or 64-bit. Possible values: `32`, `64`

Registers

Specification: This property shall contain information defining the registers exported by this Programmable Device.

Array Item Properties:

Property	Type	L	R	Specification
Identifier	NamedComponent			This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components).
Alias	ComponentAlias			This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components).
SizeBytes	integer			This property shall contain the size in bytes of this register. If this property is omitted, the register shall be presumed to be one byte in size.
Bitfields	array of Registers.Bitfields			This property shall contain the bitfields defined within this register.

Registers.Bitfields

Specification: This property shall contain the bitfields defined within this register.

Property	Type	L	R	Specification
Identifier	NamedComponent			This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components).
Alias	ComponentAlias			This property shall contain a secondary Identifier for this instance of a named

Property	Type	L	R	Specification
				component. This Identifier shall be unique across all named components within this document (excluding references to components).
Type	string			This property shall contain the type of this bitfield element.
LeastSignificantBit	integer			This property shall contain the least significant bit for this bitfield element.
SizeBits	integer			This property shall contain the size in bits of this bitfield element.
Reset	integer			This property shall contain the value of this bitfield element after a reset.
Access	string			This property shall contain the type of access intended for this bitfield element. Possible values: `ro` (read-only access: the bitfield element may be read but should not be written to), `rw` (read/write access: the bitfield element may be read or written to), `wo` (write-only: the bitfield element may be written to but should not be read), `w1c` (write one to clear: writing a 1 to a bit in the bitfield element clears that bit.)
EnumeratedValues	array of Registers.Bitfields.EnumeratedValues			This property shall contain the enumerated values for this bitfield element, if applicable.

Registers.Bitfields.EnumeratedValues

Specification: This property shall contain the enumerated values for this bitfield element, if applicable.

Property	Type	L	R	Specification
Value	integer			This property shall contain the numeric value of the enumerated option.
Name	string			This property shall contain the symbolic name of the enumerated option.
Description	string			This property shall contain a vendor-defined description of the state/code.
VendorDefined	boolean			This property shall indicate if this value falls within an OEM/vendor-reserved range.

PhysicalSignals

Specification: This property shall contain information about the physical signals supported by this Programmable Device.

Array Item Properties:

Property	Type	L	R	Specification
Identifier	NamedComponent			This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components).
Alias	ComponentAlias			This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components).
IdentifierPrefix	string			This property shall contain the portion of the identifier that is common across

Property	Type	L	R	Specification
				a set of physical signals. This prefix should include everything up to the netlist name for the associated signals. When the `PhysicalPinPrefix` and pin numbers are appended to this property, the resulting identifier shall form the complete `Identifier` for a physical signal in the set. Each physical signal identifier in the set shall be unique across all named components within this document (excluding references to components). This property is mutually exclusive with the `Identifier` property and shall not be present if `Identifier` is present, or vice versa.
FirstIdentifier	integer			This property shall contain the starting value for the identifier that is a set of physical signals.
PhysicalPin	string			This property shall contain the name of the package pin for this signal.
PhysicalPinPrefix	string			This property shall contain the portion of the physical pin name that is common across a set of physical signals. When pin numbers are appended to this property, the resulting names shall form the physical pin names for physical signals in the set. This property is mutually exclusive with the `PhysicalPin` property and shall not be present if `PhysicalPin` is present, or vice versa.
FirstPhysicalPin	integer			This property shall contain the first physical pin number for a set of

Property	Type	L	R	Specification
				physical signal pins. This property is mutually exclusive with the `PhysicalPin` property and shall not be present if `PhysicalPin` is present, or vice versa. This property shall only be present if `PhysicalPinPrefix` is also present. If this property is not present, but `PhysicalPinPrefix` is present, then the default value of 0 shall be used for `FirstPhysicalPin.
PhysicalPinCount	integer			This property shall contain the number of consecutive physical pin numbers in a set of physical signal pins. This property is mutually exclusive with the `PhysicalPin` property and shall not be present if `PhysicalPin` is present, or vice versa. This property shall only be present if `PhysicalPinPrefix` is also present
Direction	Direction			This property shall contain the direction of a signal, virtual wire, or GPIO.
DriveCapabilities	array of string			This property shall contain the output drive capabilities for this signal when configured as an output.
PowerWell	RequiredPowerState			This property shall contain the power domain of the voltage rail needed for this signal.
PullUp	boolean			This property shall indicate that this signal has a pull-up resistor. If this property is not present, a default value of `false` shall be assumed.

Property	Type	L	R	Specification
PullDown	boolean			This property shall indicate that this signal has a pull-down resistor. If this property is not present, a default value of `false` shall be assumed.
IOType	string			This property shall contain the I/O standard for this signal. Possible values: `LVCMOS12`, `LVCMOS15`, `LVCMOS18`, `LVCMOS25`, `LVCMOS33`

VirtualWires

Specification: This property shall contain information about the virtual wires supported by this Programmable Device.

Array Item Properties:

Property	Type	L	R	Specification
Identifier	NamedComponent	Y	Y	This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components).
Alias	ComponentAlias			This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components).
Start	integer	Y	Y	This property shall specify the first wire in a set of contiguous virtual wires.
TotalWidth	integer	Y	Y	This property shall specify the total number of virtual wires in a set of

Property	Type	L	R	Specification
				contiguous virtual wires.
Behavior	VirtualWires.Behavior			This property shall contain a definition for a set of virtual wires.
InternalNetlistMappingNames	array of NameSet			This property shall contain an ordered set of names.
BusMappings	array of VirtualWires.BusMappings			This property shall contain information about buses mapped to virtual wires.
RegisterMappings	array of VirtualWires.RegisterMappings			This property shall contain information about registers mapped to virtual wires.
PhysicalSignalMappings	array of VirtualWires.PhysicalSignalMappings			This property shall contain information about physical signals mapped to virtual wires.
IPBlockMappings	array of VirtualWires.IPBlockMappings			This property shall contain information about IP blocks mapped to virtual wires.
Capabilities	array of VirtualWires.Capabilities			This property shall contain capabilities that control the behavior of virtual wires.

VirtualWires.Behavior

Specification: This property shall contain a definition for a set of virtual wires.

Property	Type	L	R	Specification
DefaultState	integer	Y	Y	This property shall specify the default state for the set of virtual wires.
ActiveState	integer	Y	Y	This property shall specify the active state for the set of virtual wires.

Property	Type	L	R	Specification
Direction	Direction	Y	Y	This property shall contain the direction of a signal, virtual wire, or GPIO.
StartupPolarity	string	Y	Y	This property shall specify the initial polarity for the set of virtual wires. Possible values: `High` (A logical high means the signal is in an active state), `Low` (A logical low means the signal is in an active state), `HighImpedance` (The signal is in a logically disabled state)
RequiredPowerState	RequiredPowerState			This property shall contain the minimum power state to utilize a bus.

VirtualWires.BusMappings

Specification: This property shall contain information about buses mapped to virtual wires.

Property	Type	L	R	Specification
Alias	ComponentAlias			This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components).
Interfaces	array of string			This property shall contain the specific buses that virtual wires are being mapped to. The cardinality of this array shall match the `Width`. Each entry in this array shoall be either an `Identifier` or an `Alias` in the `Buses` section of this document.
Start	integer			This property shall specify the index of the first virtual wire being mapped to a

Property	Type	L	R	Specification
				bus. Indices are zero-based and refer to the `InternalNetlistMappingNames` defined for this set of virtual wires. If this property is not present, the start index shall be assumed to be zero.
Width	integer			This property shall specify the number of virtual wires being mapped to buses. Indices are zero-based and refer to the `InternalNetlistMappingNames` defined for this set of virtual wires. If this property is not present, the width shall be assumed to be the cardinality of the `InternalNetlistMappingNames` set.
NameSet	NameSet			This property shall reference the names of buses that the virtual wires are being mapped to. Each referenced bus name shall be either an `Identifier` or an `Alias` for an element in corresponding `Interfaces` bus in the `Buses` section of this document.
Signals	array of string			This property shall contain the set of signals that virtual wires are being mapped to virtual wires. The cardinality of this array shall match the `` `Width `` .
Description	string			This property shall contain a description of the mapping.

VirtualWires.RegisterMappings

Specification: This property shall contain information about registers mapped to virtual wires.

Property	Type	L	R	Specification
Alias	ComponentAlias			This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components).
Start	integer			This property shall specify the index of the first virtual wire being mapped to a register. Indices are zero-based and refer to the `InternalNetlistMappingNames` defined for this set of virtual wires. If this property is not present, the start index shall be assumed to be zero.
Width	integer			This property shall specify the number of virtual wires being mapped to registers. Indices are zero-based and refer to the `InternalNetlistMappingNames` defined for this set of virtual wires. If this property is not present, the width shall be assumed to be the cardinality of the `InternalNetlistMappingNames` set.
NameSet	NameSet			This property shall reference the names of registers that the virtual wires are being mapped to. Each referenced register name shall be either an `Identifier` or an `Alias` in the `Registers` section of this document.
RegionNames	array of string			This property shall reference the regions containing the registers that virtual wires are being mapped to. The referenced region names shall each be

Property	Type	L	R	Specification
				either an `Identifier` or an `Alias` for a `Regions` entry in the `MemoryMap` section of this document.
ManyToOne	boolean			This property shall indicate that a mapping applies to all virtual wires in the `InternalNetlistMappingNames` set. This property shall not contain `true` unless `Width` contains 1. If this property is not present, a default value of `false` shall be assumed.
Bitfields	array of string			This property shall reference a list of named bitfields in the register that the virtual wires are mapped to. Each referenced bitfield shall be either an `Identifier` or an `Alias` in the `Registers` section of this document.
Description	string			This property shall contain a description of the mapping.

VirtualWires.PhysicalSignalMappings

Specification: This property shall contain information about physical signals mapped to virtual wires.

Property	Type	L	R	Specification
Alias	ComponentAlias			This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components).
Start	integer			This property shall specify the index of the first virtual wire being mapped to a physical signal. Indices are zero-based

Property	Type	L	R	Specification
				and refer to the `InternalNetlistMappingNames` defined for this set of virtual wires. If this property is not present, the start index shall be assumed to be zero.
Width	integer			This property shall specify the number of virtual wires being mapped to physical signals. Indices are zero-based and refer to the `InternalNetlistMappingNames` defined for this set of virtual wires. If this property is not present, the width shall be assumed to be the cardinality of the `InternalNetlistMappingNames` set.
NameSet	NameSet			This property shall reference the names of physical signals that the virtual wires are being mapped to. Each referenced physical signal name shall be either an `Identifier` or an `Alias` in the `PhysicalSignals` section of this document.
Description	string			This property shall contain a description of the mapping.

VirtualWires.IPBlockMappings

Specification: This property shall contain information about IP blocks mapped to virtual wires.

Property	Type	L	R	Specification
Alias	ComponentAlias			This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components

Property	Type	L	R	Specification
				within this document (excluding references to components).
Start	integer			This property shall specify the index of the first virtual wire being mapped to an IP block. Indices are zero-based and refer to the `InternalNetlistMappingNames` defined for this set of virtual wires. If this property is not present, the start index shall be assumed to be zero.
Width	integer			This property shall specify the number of virtual wires being mapped to IP blocks. Indices are zero-based and refer to the `InternalNetlistMappingNames` defined for this set of virtual wires. If this property is not present, the width shall be assumed to be the cardinality of the `InternalNetlistMappingNames` set.
ManyToOne	boolean			This property shall indicate that a mapping applies to all virtual wires in the `InternalNetlistMappingNames` set. This property shall not contain `true` unless `Width` contains 1. If this property is not present, a default value of `false` shall be assumed.
NameSet	NameSet			This property shall specify the number of virtual wires in a set that are being mapped to IP blocks. Each referenced IP block shall be either an `Identifier` or an `Alias` in the `IP blocks` section of this document.

Property	Type	L	R	Specification
Description	string			This property shall contain a description of the mapping.

VirtualWires.Capabilities

Specification: This property shall contain capabilities that control the behavior of virtual wires.

Property	Type	L	R	Specification
NameSet	NameSet			This property shall specify the virtual wires to which the capability flags apply. Each named virtual wire shall be described by the `InternalNetlistMappingNames` for this mapping.
Bitmask	integer			This property shall contain the capabilities bitmask for a virtual wire.
MaskType	string			This property shall contain the type of bitmask that applies to this virtual wire as defined in the relevant specification. Possible values: `FlexIO`, `Sideband`, `OEM`

Buses

Specification: This property shall contain information about the logical and physical buses connected to this Programmable Device.

Array Item Properties:

Property	Type	L	R	Specification
BusType	string	Y	Y	This property shall contain the type for this bus. Possible values: `DisplayPort`, `ESPI`, `I2C`, `I3C`, `JTAG`, `LTPI`, `MPESTI`, `NCSI_RBT`, `PCIe`, `PECI`,

Property	Type	L	R	Specification
				`QSPI`, `SGMII`, `SPI`, `SGPIO`, `UART`, `USB`, `OEM`
Identifier	NamedComponent	Y	Y	This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components).
Alias	ComponentAlias			This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components).
RequiredPowerState	RequiredPowerState			This property shall contain the minimum power state to utilize a bus.
SPDM	boolean			This property shall indicate whether an SPDM connection can be established over this bus. If this property is omitted, it shall be assumed that an SPDM connection cannot be established over this bus.
RandomAccessCapable	boolean			This property shall indicate whether this bus supports random access into the memory space of the Programmable Device.
HostAccessible	boolean			This property shall indicate whether this bus is accessible to the host.
MaxClockSpeedMHz	number			This property shall indicate the maximum clock speed this bus supports, in MHz.

Property	Type	L	R	Specification
DataRateMultiplier	integer			This property shall indicate a multiplication factor for the data rate for a bus. If this property is omitted, a default value of 1 shall be assumed.
AlertSignal	string			This property shall indicate the alias or netname for the alert signal associated with this bus.
Tunneled	Buses.Tunneled			This property shall contain information about how a bus is tunneled. This property shall be omitted if the bus is not tunneled.
IPBlockMappings	array of string			This property shall indicate the IP block that implements support for this bus. The referenced IP block shall be either an `Identifier` or an `Alias` in the `IPBlocks` section of this document.
Signals	array of Buses.Signals			This property shall contain information about the signals associated with a bus.
Tunneling	Buses.Tunneling			This property shall provide tunneling information for the bustunneling.
MPESTI	Buses.MPESTI			This property shall provide MPESTI-specific information for the bus. This property shall be omitted if `BusType` is not `MPESTI`.
SGPIO	Buses.SGPIO			This property shall provide SGPIO-specific information for the bus. This property shall be omitted if `BusType` is not `SGPIO`.
ESPI	Buses.ESPI			This property shall provide ESPI-specific information for the bus. This

Property	Type	L	R	Specification
				property shall be omitted if `BusType` is not `ESPI`.
SPI	Buses.SPI			This property shall provide SPI-specific information for the bus. This property shall be omitted if `BusType` is not `SPI`.
QSPI	Buses.QSPI			This property shall provide QSPI-specific information for the bus. This property shall be omitted if `BusType` is not `QSPI`.
OEM	Buses.OEM			This property shall provide OEM-specific information for the bus. This property shall be omitted if `BusType` is not `OEM`.

Buses.Tunneled

Specification: This property shall contain information about how a bus is tunneled. This property shall be omitted if the bus is not tunneled.

Property	Type	L	R	Specification
Type	string			This property shall indicate the type of bus over which this bus is tunneled. Possible values: `LTPI`, `OEM`
Interface	string			This property shall indicate the specific interface over which this bus is tunneled. The referenced interface shall be either an `Identifier` or an `Alias` in the `Buses` section of this document.

Buses.Signals

Specification: This property shall contain information about the signals associated with a bus.

Property	Type	L	R	Specification
PhysicalPin	string			This property shall indicate the physical pin associated with this signal.
Alias	string			This property shall contain an alias for this signal.
NetName	string			This property shall contain the net name for this signal.

Buses.Tunneling

Specification: This property shall provide tunneling information for the bustunneling.

Property	Type	L	R	Specification
GPIOs	GPIOs			This property shall contain information about a set of GPIOs.
TunneledBuses	array of Buses.Tunneling.TunneledBuses			This property shall contain an enumeration of buses tunneled through this LTPI bus.
DataChannelMappedRegions	array of Buses.Tunneling.DataChannelMappedRegions			This property shall contain the regions of memory mapped in the data channel.

Buses.Tunneling.TunneledBuses

Specification: This property shall contain an enumeration of buses tunneled through this LTPI bus.

Property	Type	L	R	Specification
BusType	string			This property shall contain the type of the tunneled bus, as indicated in the `BusType` property in the `Buses` section of this document.

Property	Type	L	R	Specification
BusId	string			This property shall identify a tunneled bus. The referenced bus shall be either an `Identifier` or an `Alias` in the `Buses` section of this document.

Buses.Tunneling.DataChannelMappedRegions

Specification: This property shall contain the regions of memory mapped in the data channel.

Property	Type	L	R	Specification
Identifier	NamedComponent			This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components).
Alias	ComponentAlias			This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components).
OffsetBytes	integer			This property shall contain the offset in memory to the beginning of this mapped memory region. This property may be omitted if `OffsetBytesHex` is present.
OffsetBytesHex	string			This property shall contain the offset in memory to the beginning of this mapped memory region, in hexadecimal. This property may be omitted if `OffsetBytes` is present.
SizeBytes	integer			This property shall contain the size in bytes for this mapped memory region.

Property	Type	L	R	Specification
				This property may be omitted if `SizeBytesHex` is present.
SizeBytesHex	string			This property shall contain the size in bytes for this mapped memory region, in hexadecimal. This property may be omitted if `SizeBytes` is present.
MappedMemoryRangeStartHex	string			This property shall contain offset in the Programmable Device memory map for the beginning of the data channel mapped memory region, in hexadecimal. This property shall be omitted if `MappedMemoryRegion` is present.
MappedMemoryRegion	string			This property shall indicate a memory region that is mapped to the data channel. The referenced region shall be either an `Identifier` or an `Alias` for a `Region` in the `MemoryMap` section of this document. This property shall be omitted if `MappedMemoryRangeStartHex` is present.
Description	string			This property shall describe a memory region that is mapped to the data channel.

Buses.MPESTI

Specification: This property shall provide MPESTI-specific information for the bus. This property shall be omitted if `BusType` is not `MPESTI`.

Property	Type	L	R	Specification
ExternalMUXSupported	boolean			This property shall indicate whether this MPESTI bus supports an external mux.
MaximumTargetsSupported	integer			This property shall indicate the maximum number of targets supported by this MPESTI bus.
ConnectedToTarget	boolean			This property shall indicate whether this bus is connected to an MPESTI target.
PayloadBufferDestructive	boolean			This property shall be true if the discovery payload mailbox is a single destructive flat buffer holding only the most recently discovered target; false/omitted if it is a non-destructive per-target mailbox window.
InitiatorsPerIPBlock	integer			This property shall contain the number of MPESTI initiators (PESTI wires) instances implemented within this IP block.
InitiatorBlockSetIdentifier	string			This property shall contain the identifier grouping this IP block into an Initiator IP Block Set.
InitiatorBlockSetCount	integer			This property shall contain the total number of Initiator IP Block Sets (Initiator Super Set size) supported by the Programmable Device.
OperationalMode	string			This property shall contain the operational mode of the M-PESTI MUX/Switch. Possible values: `Target`, `Snoop`
FanoutMethod	string			This property shall contain the method used to select/switch among M-PESTI

Property	Type	L	R	Specification
				channels. Possible values: `MUX`, `Switch`

Buses.SGPIO

Specification: This property shall provide SGPIO-specific information for the bus. This property shall be omitted if `BusType` is not `SGPIO`.

Property	Type	L	R	Specification
GPIOs	GPIOs	Y	Y	This property shall contain information about a set of GPIOs.

Buses.ESPI

Specification: This property shall provide ESPI-specific information for the bus. This property shall be omitted if `BusType` is not `ESPI`.

Property	Type	L	R	Specification
DummyBytes	integer			This property shall define the number of dummy padding bytes for the ESPI interface.

Buses.SPI

Specification: This property shall provide SPI-specific information for the bus. This property shall be omitted if `BusType` is not `SPI`.

Property	Type	L	R	Specification
DummyBytes	integer			This property shall define the number of dummy padding bytes for the SPI interface.

Buses.QSPI

Specification: This property shall provide QSPI-specific information for the bus. This property shall be omitted if `BusType` is not `QSPI`.

Property	Type	L	R	Specification
DummyBytes	integer			This property shall define the number of dummy padding bytes for the QSPI interface.

Buses.OEM

Specification: This property shall provide OEM-specific information for the bus. This property shall be omitted if `BusType` is not `OEM`.

No properties defined.

Common Definitions

ComponentAlias

Specification: This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components).

Type: string

Direction

Specification: This property shall contain the direction of a signal, virtual wire, or GPIO.

Type: string

Allowed Values: "input", "output", "bidir"

GPIOs

Specification: This property shall contain information about a set of GPIOs.

Array Item Properties:

Property	Type	L	R	Specification
Identifier	NamedComponent			This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components).
Alias	ComponentAlias			This property shall contain a secondary Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding references to components).
Latency	string			This property shall contain the latency for this GPIO. Possible values: `Low`, `Normal`
StartBit	integer			This property shall contain the start bit for this GPIO.
WidthBits	integer			This property shall contain the width in bits for this GPIO.
VirtualWireEndpoint	string			This property shall indicate the virtual wire attached to this GPIO. The referenced virtual wire shall be either an `Identifier` or an `Alias` in the `VirtualWires` section of this document.
Description	string			This property shall describe the intended use for this GPIO.
Direction	Direction			This property shall contain the direction of a signal, virtual wire, or GPIO.

Property	Type	L	R	Specification
ByteOffset	integer			This property shall contain the byte offset for this GPIO.

NameSet

Specification: This property shall contain an ordered set of names.

Property	Type	L	R	Specification
Count	integer			This property shall define the number of names in the set.
List	array of string			This property shall contain an explicit list of names. The cardinality of the list shall match `Count`.
Block	NameSet.Block			This property shall provide a block definition of a set of names.

NameSet.Block

Specification: This property shall provide a block definition of a set of names.

Property	Type	L	R	Specification
NamePrefix	string			This property shall contain a name prefix for a subset of names.
NameOffset	integer			This property shall contain an offset within a numerical suffix for a set of names. The mapping for the Nth name (zero-based) in the set of names shall be constructed by appending (`NameOffset` + N) to `NamePrefix`.

NamedComponent

Specification: This property shall contain an Identifier for this instance of a named component. This Identifier shall be unique across all named components within this document (excluding

references to components).

Type: string

ReferencedBusList

Specification: This object shall contain a collection of references to hardware or logical buses.

Array Item Properties:

Property	Type	L	R	Specification
Type	string			This property shall contain the type of the bus being referenced.
Identifier	string			This property shall contain the name of the instance of the bus type being referenced.
SpecBusName	string			This property shall contain the name used in the spec to reference this bus.

RequiredPowerState

Specification: This property shall contain the minimum power state to utilize a bus.

Type: string

Allowed Values: "G3", "Pre-STBY", "STBY", "S5", "S0", "Other"

SpecReference

Specification: This property shall contain a reference to an OCP specification.

Property	Type	L	R	Specification
Name	string			This property shall contain the name of the OCP specification being referenced.
Version	string			This property shall contain the version of the OCP specification being

Property	Type	L	R	Specification
				referenced.
Revision	string			This property shall contain the revision of the OCP specification being referenced.